NOTE

Best practice when implementing digital twins is to start with a small pilot project rather than a full-scale implementation. This approach allows you to validate your ideas and gain stakeholder buy-in with minimal risk and investment upfront.

1.5.2 Data quality

It will come as no surprise that creating an effective digital representation of a physical object or system depends on having accurate, complete, and consistent data about the real world. Yet, although modern organizations are reportedly drowning in data, poor data quality sabotages even the most sophisticated digital twins. As Herbert Simon observed, "a wealth of information creates a dearth of attention", and this abundance often masks underlying quality issues that can undermine digital twin effectiveness.

INCOMPLETE AND UNRELIABLE DATA

Unless you are designing and building your system from scratch, be it a new building, electric vehicle, or iron ore mine, you will likely be dealing with aged assets and infrastructure with varying levels of instrumentation and information available. Just getting operational data streams from your OT network into the IT space where your digital twin will be deployed can be a significant challenge, and the resulting data is often both incomplete and of questionable quality.

Data quality issues can arise from sensors drifting out of calibration over time, connectivity problems that create gaps in time-series data, and equipment failures introducing noise and anomalies. The data that is available could use different measurement standards, communication protocols, and sampling frequencies, that make it difficult to achieve the consistency and accuracy that effective digital twins require. Some of the data may be locked away behind vendor contracts that do not even give you access to data from your own equipment.